

FA 5

Problem 1 (5 points)

An experiment consists of drawing three cards in succession from a well-shuffled ordinary deck of cards. Let E_1 be a king on the first draw, E_2 a king on the second, and E_3 a king on the third.

- (a) $P(\overline{E_1} \overline{E_2})$: The probability that a king is not drawn on the first draw and a king is not drawn on the second draw.
- (b) $P(E_1 + E_2)$: The probability that a king is drawn on the first draw, the second draw, or both.
- (c) $E_1 + E_2$: The event that a king occurs on the first draw or the second draw.
- (d) $P(E_3 | E_1 \overline{E_2})$: The conditional probability that a king is drawn on the third draw, given that a king was drawn on the first draw and no king was drawn on the second.
- (e) $\overline{E_1} \overline{E_2} \overline{E_3}$: The event that no king is drawn in any of the three draws.
- (f) $P(E_1 E_2 + \overline{E_2} E_3)$: The probability that kings are drawn on the first and second draws, or a king is drawn on the third draw without a king on the second.

Problem 2 (5 points)

Three dice are rolled and X is the sum of the three upper faces.

a. Probability Distribution of X

```
dice_grid <- expand.grid(d1 = 1:6, d2 = 1:6, d3 = 1:6)
sums <- rowSums(dice_grid)
dist_table <- table(sums) / length(sums)
knitr::kable(dist_table, col.names = c("Sum (X)", "P(X)"))
```

Sum (X)	P(X)
3	0.0046296
4	0.0138889
5	0.0277778
6	0.0462963
7	0.0694444
8	0.0972222
9	0.1157407
10	0.1250000
11	0.1250000
12	0.1157407

Sum (X)	P(X)
13	0.0972222
14	0.0694444
15	0.0462963
16	0.0277778
17	0.0138889
18	0.0046296

b. Find $P(7 \leq X \leq 11)$

```
prob_range <- sum(dist_table[names(dist_table) %in% 7:11])
print(prob_range)
```

```
## [1] 0.5324074
```

Problem 3 (5 points)

[cite_start][cite: 28] Given the following probability distribution[cite: 29, 34]:

X	-10	-20	30
$p(X)$	1/5	3/10	1/2

```
x_vals <- c(-10, -20, 30)
p_vals <- c(1/5, 3/10, 1/2)

# (a) E(X)
ex <- sum(x_vals * p_vals)

# (b) E(X^2)
ex2 <- sum(x_vals^2 * p_vals)

# (c) E[(X - X_bar)^2]
var_x <- sum((x_vals - ex)^2 * p_vals)

# (d) E(X^3)
ex3 <- sum(x_vals^3 * p_vals)
```

- (a) Expectation $E(X)$: 7
- (b) Expectation $E(X^2)$: 590
- (c) Variance $E[(X - \bar{X})^2]$: 541
- (d) Expectation $E(X^3)$: 1.09×10^4

Problem 4 (5 points)

Seating 3 men and 3 women at a round table[cite: 39]:

- (a) **No restriction?**[cite: 40]: $(6 - 1)! = 5! = 120$.
- (b) **Two particular women must not sit together?**[cite: 41]: Total - (Women together) = $5! - (4! \times 2) = 72$.
- (c) **Each woman between two men?**[cite: 42]: $(3 - 1)! \times 3! = 2 \times 6 = 12$.

Problem 5 (5 points)

Selecting from 6 men, 8 women, 4 boys, and 5 girls[cite: 45]:

- (a) **No restrictions (Selecting 2M, 4W, 3B, 3G)?**

```
ans_5a <- choose(6, 2) * choose(8, 4) * choose(4, 3) * choose(5, 3)
ans_5a
```

```
## [1] 42000
```

- (b) **Particular man and woman must be selected?**

```
ans_5b <- choose(5, 1) * choose(7, 3) * choose(4, 3) * choose(5, 3)
ans_5b
```

```
## [1] 7000
```

Problem 6 (5 points)

[Prove that for any events E_1 and E_2 , $P(E_1 + E_2) \leq P(E_1) + P(E_2)$.]

Proof: Using the Addition Rule of Probability[cite: 51]:

$$P(E_1 + E_2) = P(E_1) + P(E_2) - P(E_1 E_2)$$

Since $P(E_1 E_2) \geq 0$ (probability cannot be negative), subtracting it from the sum $P(E_1) + P(E_2)$ will always result in a value less than or equal to that sum:

$$P(E_1) + P(E_2) - P(E_1 E_2) \leq P(E_1) + P(E_2)$$

Thus:

$$P(E_1 + E_2) \leq P(E_1) + P(E_2)$$